

A finite certificate for near alternating sign matrices with weight multisets through 6×6

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Abstract

A near alternating sign matrix has entries in $\{0, 1, -1\}$, with alternating nonzero signs in every row and column. We give an exact finite certificate that every pair of positive row- and column-weight multisets satisfying the Gale–Ryser conditions is realized by such a matrix whenever both dimensions are at most six. The certificate consists of 15,583 explicit matrices, including 7,788 for dimension 6×6 . A standalone integer verifier regenerates the entire admissible domain and checks the entries, alternation, weight multisets, and exact coverage. A separately implemented verification confirms these checks. Row and column weight orderings are chosen independently; fixed-order existence is not asserted. This proves only the stated finite restriction of Traetta’s multiset conjecture. The unbounded conjecture remains unresolved by this work.

1 Statement and context

For an $m \times n$ matrix A , its row and column *weights* are the numbers of nonzero entries in its rows and columns. A *near alternating sign matrix* (NASM) is a matrix in $\{0, 1, -1\}^{m \times n}$ such that, after zeros are deleted from any row or column, consecutive entries have opposite signs. Either initial sign is allowed; a single nonzero satisfies the alternation condition.

For a finite sequence h , write $h^\downarrow$ and $h^\uparrow$ for its nonincreasing and nondecreasing rearrangements. The precise finite statement proved here is the following.

Theorem 1. *Let $1 \leq m, n \leq 6$ be integers, and let*

$$h \in \{1, \dots, n\}^m, \quad k \in \{1, \dots, m\}^n.$$

Suppose that

$$\sum_{i=1}^m h_i = \sum_{j=1}^n k_j, \quad \sum_{i=1}^t h_i^\downarrow \leq \sum_{j=1}^n \min(t, k_j) \quad (1 \leq t \leq m). \quad (1)$$

There exist a permutation h' of h , an independently chosen permutation k' of k , and an $m \times n$ NASM whose row-weight sequence is h' and whose column-weight sequence is k' .

The hypotheses in (1) are the Gale–Ryser conditions for a binary matrix with the specified weights. They are necessary for a NASM because taking absolute values preserves all weights; see [1, Lemma 1.3]. The unrestricted positive-weight multiset assertion is Conjecture 2.5 of that source. Its Theorem 2.3 characterizes fixed-order NASM existence by a binary support whose graph, joining

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consecutive nonzeros in rows and columns, is bipartite. Theorem 2.7 gives a sufficient convex-support condition, and Section 3 supplies composition constructions. Our contribution relative to the inspected version is a complete, directly checkable certificate for the domain $1 \leq m, n \leq 6$; no such complete finite enumeration appears in that text. We make no claim of literature-wide priority.

The distinction between multisets and prescribed sequences is essential. Traetta’s Example 2.4 shows that the fixed row and column sequences $(3, 2, 3)$ are not simultaneously realizable, although their multisets are realized by

$$\begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 0 \end{pmatrix}, \quad h' = k' = (3, 3, 2).$$

Theorem 1 permits these independently selected weight orders. It does not rely on rearranging the rows or columns of an already signed matrix, an operation that need not preserve alternation.

As adjacent computational methodology, exact finite certification of matrix-existence statements also occurs in Zeng’s work on the sparse-complement two-eigenvalue conjecture [2]. That comparison concerns finite enumeration and explicit matrix certificates; it supplies no NASM existence result used here.

2 Certificate and exhaustive verification

The proof combines elementary normalization with exact verification of a finite certificate. The supplementary corpus contains one signed matrix for each admissible key

$$(m, n, h^\uparrow, k^\uparrow).$$

Each record stores the dimensions, the two nondecreasing weight tuples, and the complete signed matrix. The witness matrix retains its actual row and column orders. Table 1 gives the number of admissible keys for each ordered dimension pair; all 36 pairs are enumerated, without a transposition reduction.

Table 1: Admissible positive weight-multiset pairs, each with one verified NASM witness. Rows index m and columns index n .

$m \backslash n$	1	2	3	4	5	6
1	1	1	1	1	1	1
2	1	3	5	8	11	15
3	1	5	15	33	64	112
4	1	8	33	103	260	579
5	1	11	64	260	843	2323
6	1	15	112	579	2323	7788

Computer-assisted proof of Theorem 1. Both the common total and (1) depend only on the input multisets. Thus each admissible ordered input determines a unique canonical key. This normalization acts on input weights, not on a signed matrix.

For fixed m, n , generate every nondecreasing tuple h of length m with entries in $\{1, \dots, n\}$, and every nondecreasing tuple k of length n with entries in $\{1, \dots, m\}$. One exhaustive generator chooses

its next entry from the current lower bound to the upper bound, and recurses with the chosen value as the next lower bound. The empty tuple is the terminal case. Induction on tuple length shows that each permitted tuple is generated exactly once. Retain precisely the pairs satisfying (1), evaluating the left side as the sum of the last t entries of the nondecreasing h . Let D denote the resulting set of keys over all 36 dimension pairs.

Independently read every certificate record. Require integer dimensions in the stated bounds, canonical positive weight tuples in their bounds, and a matrix of exactly the specified shape with integer entries in $\{0, 1, -1\}$. For every actual row and column, delete zeros and check that each consecutive pair sums to zero. Count the nonzeros in each row and column, sort these counts, and compare them with the stored input tuples. These tests are exactly the NASM and weight-multiset requirements. Reject duplicate keys, and let W be the set of checked witness keys.

The completed exact computation gives

$$|D| = |W| = 15,583, \quad D \setminus W = W \setminus D = \emptyset. \quad (2)$$

In particular, it checks set equality, not merely equality of cardinalities. The detailed counts are in Table 1. Every element of D therefore has an explicit matrix passing all the preceding tests.

Finally, take any ordered input in the theorem. Its canonical key belongs to D and hence has a witness. Equality of the row-weight multisets gives a bijection between occurrences in h and the actual row weights of this matrix, including repeated values. This bijection gives the required permutation h' . Independently, equality of the column-weight multisets gives k' . The matrix already alternates in its stored order and has these actual weight sequences. This establishes every quantified case of the theorem. $\square$

The accompanying verifier implements these checks using Python integers. It considers 371,371 positive canonical multiset pairs, of which 20,731 have equal totals. For each equal-total pair it also computes an integral maximum flow in the network with source-to-row capacities h_i , row-to-column capacities one, and column-to-sink capacities k_j . A flow of value $\sum_i h_i$ corresponds to a binary support with those weights. This test agrees with the direct Gale–Ryser test on all 20,731 pairs; any disagreement aborts verification. The flow check is a redundant encoding check, not an assumption about NASM existence.

A separate implementation regenerated the domain by enumerating all ordered tuples, sorting, and deduplicating, with the tuple counts checked against $\binom{m+n-1}{m}$ and $\binom{m+n-1}{n}$. It evaluated the inequality for every subset of rows, using the fact that the largest sum over subsets of size t is the sum of the t largest row weights. Across 1,106,840 subset tests, the resulting admissibility decisions agreed with the row-prefix and transposed column-prefix formulations. Direct verification of all matrices again found 15,583 valid witnesses, zero missing or extra keys, and zero duplicates. This second implementation imports no code from the witness search or the first verifier.

Additional implementation controls exhaustively enumerated all 21,297 ternary matrices across the nine ordered dimension pairs with $m, n \leq 3$. Their positive NASM weight keys agree with the 33 admissible keys in that range. Eleven corruption controls were rejected, including altered signs, invalid entries or shape, wrong weights, and deleted or duplicated records. These controls test the implementation; the full certificate check and domain equality in (2) remain the decisive computation.

3 Reproduction and limits

The supplementary archive contains the full line-delimited JSON certificate and the standalone standard-library verifier. After extraction, run the following command from the archive’s root:

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python3 evidence/verify_f6.py --report verification.json
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The report records all dimension counts and the coverage checks. A failed matrix test, duplicate, domain mismatch, or disagreement between the two admissibility tests raises an error. The certificate is finite data that must be checked together with the argument above; the reported counts alone do not constitute its verification.

The witnesses were found by a C++17 search that fills rows from top to bottom. For each remaining row weight, it tries support subsets and their two alternating signings, tracking the last nonzero sign and nonzero count in each column. Columns retain their actual positions, and their final weights must form the target multiset. The final run found all witnesses without reaching its node cap. Neither the correctness nor the completeness of this search is required by the positive certificate proof: the verifier uses the final matrices and independently generates the complete domain. All decisive arithmetic is integral; floating-point runtime fields in the search records are not mathematical evidence.

The result includes one-row and one-column cases and permits either initial nonzero sign. It covers positive weights only, with both dimensions at most six. No claim is made for zero-weight lines, larger dimensions, prescribed weight orders, or a general construction. The unbounded conjecture remains unresolved by this work. The corpus may be used to test proposed general constructions, but success on it cannot establish an unbounded assertion.

The source comparison is restricted to the inspected version of [1]. Current literature-wide duplication and priority have not been established, and no priority claim is made.

Computational and AI assistance. OpenAI Codex assisted with the mathematical exposition, generation of the search and verification programs, execution of the computations, and preparation of this manuscript. A separate Codex review context wrote and ran the second verifier described above. This is a separate implementation and review pass, not external peer review or a proof-assistant formalization. The mathematical claim is supported by the explicit certificate, the exhaustive verification, and the finite-domain argument given here.

References

- [1] T. Traetta, *A note on near alternating sign matrices with prescribed row and column weights*, arXiv:2609.07118v1, 2026. <https://arxiv.org/abs/2609.07118v1>.
- [2] Zijian Zeng, *Exact Finite Certification of The Sparse-Complement Two-Eigenvalue Conjecture at Order Nine*, 2026. <https://doi.org/10.2139/ssrn.7379578>.